

Bursar dashboard: workflow audit

Huchu schools vertical audit · Bursar dashboard: workflow audit · bursar/workflows.md · rendered 2026-09-15

Surface: the fees and finance area of the school workspace. Pages `app/schools/finance/**` (overview by year group, class view, ledger with invoices, receipts, credits, refunds, waivers and structures, arrears) and `app/schools/fees` (redirect), the fee sections of `app/schools/documents` and `app/schools/imports`, and portal invites from guardian records. APIs `app/api/v2/schools/fees/**`, `app/api/v2/schools/finance/*`, `reports/{arrears,collections,export}`. Domain logic `app/api/v2/schools/fees/_helpers.ts`, `lib/schools/fees-v2.ts`, `lib/schools/money.ts`, `lib/schools/fiscalisation.ts`, `lib/schools/audit.ts`. Persona: BURSAR (`lib/platform/personas.ts:106-124`), plus `SCHOOL_ADMIN` and tenant admins.

Audit date: 2026-09-14. Static read of `main`, cross-checked against the roadmap (Iteration 2, S-4.6, S-5.1, S-7.3, S-7.4, S-8.2, S-9.3), the fees phase 3 spec, the marketing promises, and the K-12 benchmark §9. UI companion: `ui-ux.md` in this folder.

1. Verdict in one paragraph

The money core is the strongest part of the schools vertical. Every money verb is wired end to end, transactional, row-locked, `Decimal(14,2)`, dual-currency, DB-constrained, audited, posted to the general ledger with an idempotency key, and fiscalisable through ZIMRA. The tests are real (fee money, credit and refund, audit, posting, fiscalisation, clone). What is wrong is around the edges and in the role model. A bursar cannot send a fee reminder or answer a parent's fee question, because both go through a route that only the head may call. A draft receipt can never be posted. One bursar can create, approve and apply a scholarship alone. GL posting runs after the transaction commits and a failure is only logged. A fiscalised receipt can be voided locally with no reversal. Draft invoices take allocations. Beyond that, the functions a bursar's office runs a term on are absent: payment plans, sibling and scholarship schemes as entities, bank reconciliation of receipts, a cash-up, a statement run, dunning, and any online payment.

Most of the role and control list is now closed. The bursar holds `notify-families` and `reply`, so she can chase the arrears her own screen lists and answer a family in the office inbox; the draft receipt is gone, because a counter receipt posts when the money is taken; a waiver approved by the person who raised it cannot be applied by them; a receipt ZIMRA has accepted cannot be voided locally; and allocation, waiver and write-off all want an issued bill. GL posting still runs after the transaction commits. What changed is that its outcome is now written on the document rather than logged and lost, and the migration that backfilled it found 120 issued invoices on the seeded tenant with no ledger entry at all — the defect the audit could only predict, now a list somebody can be given. Nothing in the last sentence of the paragraph above has moved: payment plans, schemes, reconciliation, a cash-up, a statement run, dunning and online payment are all still absent, and so is the channel every one of them would need.

Runtime check (14 September 2026)

Verified on the seeded St Marys tenant as `bursar@stmarys.test` (see `../reference/runtime-verification.md`). Confirmed at runtime: B1 (both `POST /notices` and `POST /messages` return 403 "Your role cannot create reports" while the "Send reminders" button is enabled), B2 (a receipt saved with `postNow:false` is stored as DRAFT, there is no post route, and allocate and void both refuse it), B3 (one bursar created a

waiver that landed as APPLIED with the approver equal to the creator), B7 (guardian-link PATCH returns 200 for BURSAR), B8 (opening-balance import refused with "Your role cannot create students"), and the phone ledger with no invoice on the first screen. Corrected: B10, the ageing surfaces agree on this data.

Revised on 15 September 2026 against the eighteen commits that followed the audit (`01de8a3...HEAD`). A finding the work closed keeps its description and gains a line saying it is closed and what closed it; a finding the work did not reach stands as it was written; a finding the implementation proved wrong says so and says what is actually true. The runtime paragraph above is left as it was recorded, including its last sentence, which is wrong: agreeing on the seeded data is not the same as agreeing, and the four computations behind it did not. B10 carries the correction.

2. Docs versus code

Claim	Source	Code reality	Verdict
S-2.1 to S-2.8 money correctness: done	roadmap	Decimal columns, currency and base amount, source types, partial unique index on live invoices, surplus to credit, refunds, fiscalisation, six audited actions. All confirmed in <code>_helpers.ts</code> and the fee tests.	True.
S-0.6 receipt posts to GL inline	roadmap	Posting is emitted after the receipt transaction commits (<code>receipts/route.ts:433</code> then <code>:588</code>); a FAILED posting is logged and returned as <code>accounting.accountingStatus</code> with nothing persisted on the receipt. Since revised (<code>44c4460</code>): each of the four fee documents carries <code>accountingStatus</code> , <code>journalEntryId</code> and, on a failure, the reason, behind a <code>companyId</code> -scoped index. The emit is still after the commit, deliberately — a throw there would report a good receipt as failed.	"Inline" is still not "inside the transaction", but a failure is now recorded rather than lost. Partial.
S-2.6 refunds replace the 501 stub	roadmap	<code>SchoolFeeRefund</code> with request, pay, cancel; checks available credit.	True.
S-4.6 fees by year group, ledger, clone, <code>/schools/fees</code> redirect	roadmap	All present.	True.

Claim	Source	Code reality	Verdict
S-5.1 branded invoice, receipt, statement	roadmap	Document sources exist. Receipt and statement have no page in <code>/schools/documents</code> (only invoice, report card, class list, register are surfaced, <code>school-documents-content.tsx:770-773</code>). Statement prints from the student record bar; receipt from the ledger row. Unchanged on revision: the screen still carries four views and receipt and statement are not among them. Two related things did change — the render route now serves a portal caller, so a family can open the invoice, receipt and statement written for them (B11), and the report card on the same screen turns out not to print at all unless a publish window is open, which no seeded school has (head audit B14).	True at the engine, partial at the surface. No batch statement run.
Marketing: "arrears list the bursar can work through daily"	<code>site-data.ts</code>	Arrears and ageing exist. "Send reminders" on the finance overview is enabled for <code>fees:create</code> but posts to <code>/notices</code> which requires <code>reports:create</code> . The bursar gets a 403 after composing. On the arrears page the button is correctly disabled for BURSAR. Fixed at the route in <code>e5b76a5: POST /notices takes schools.reports.notify-families</code> , which the bursar holds, so the finance overview button now works end to end. The arrears page is not fixed: it still asks <code>schools.reports.create</code> for both the bar primary and the row verb, so the bursar is refused by the screen the audit said she should chase from (B13).	Half delivered. Still in-app only.
Marketing: fiscal receipts for "fees, tuck shop, and uniform sales"	<code>SCHOOL_ADD_ONS</code>	Fee receipts only. No tuck shop or uniform sale exists in the pack.	Overclaim.
Marketing: Premier includes "Accounting, AR/AP, and banking"	pricing bands	Schools post headlessly with no accounting UI entitlement; banking reconciliation UI is parked under the scope trim.	Overclaim for schools tenants.

Claim	Source	Code reality	Verdict
Marketing: offline receipting	site FAQ	S-8.2 <code>todo</code> ; nothing in the fees UI uses the offline runtime.	Overclaim.
S-7.3 and S-7.4 portal payments and reconciliation: <code>todo</code>	roadmap	Absent. <code>lib/payments/</code> serves subscription billing only.	Accurate.
Fees phase 3 spec: accounting events PENDING for later posting	phase 3 spec	Replaced by direct posting.	Spec superseded; retire it.
Bursar dashboard drawn in <code>design/campus/leadership/BursarDashboard.dc.html</code>	design canvas	No page implements it; <code>/schools/finance</code> is the year-group picker. No roadmap story.	Design without a story.

3. Workflow inventory

Status key: Implemented = UI, API, DB, permission check and audit where sensitive; Partial = works with a named gap; Broken = the bursar hits a refusal or dead end; Missing = absent.

Setup

#	Workflow	Status	Evidence and gap
F1	Fee structures: create, edit, activate, archive, clone up the ladder	Implemented	Creation is the one structure verb not audited (<code>fees/structures/route.ts</code> POST). Fixed in <code>b21f6f2: schools.fee.structure.created</code> is written on the transaction client alongside the create.
F2	Fee structure lines (tuition, boarding, transport, levies)	Implemented	Transport and library charges are computed elsewhere and never become lines (see F22).
F3	Discounts: sibling, staff child, early payment	Missing	Only per-invoice waivers with a <code>waiverType</code> . No scheme, no automatic application.
F4	Scholarship and bursary schemes, sponsor billing (BEAM)	Missing	<code>SchoolFeeWaiver.waiverType = SCHOLARSHIP</code> per invoice only.

#	Workflow	Status	Evidence and gap
F5	Opening balances import	Partial	Requires <code>students:create</code> and <code>fees:create</code> together (<code>imports/_guard.ts:19-24</code>); only SCHOOL_ADMIN holds both. The UI lets a bursar start a job the API will refuse. Cross-currency balances import at rate 1 and are flagged. The guard is fixed in <code>50df723</code> : each entity type is checked against the one grant it belongs to, so fee structures and opening balances ask <code>schools.fees</code> and the roll asks <code>schools.students</code> , and the bursar can load what the audit found her refused from. The rate-1 cross-currency import is unchanged, which is what keeps this Partial.

Billing

#	Workflow	Status	Evidence and gap
F6	Create invoice, edit or discard while draft, issue	Implemented	Issue posts <code>SCHOOL_FEE_INVOICE_ISSUED</code> .
F7	Bulk generate per term and structure, skip existing	Implemented	Partial unique index closes the two-bursars race; one audit row per run. No preview of "this will raise 20 invoices totalling \$9,800".
F8	Issue all drafts in one action	Missing	Bulk generate creates; issuing 120 drafts is 120 row actions.
F9	Write off	Implemented	No DRAFT guard; GL posted after commit. The DRAFT guard is in (<code>b21f6f2</code>): a bill never asked for is discarded rather than written off, because there is no issue journal for the write-off to reverse. Posting is still after the commit, and now records its outcome (<code>44c4460</code>).
F10	Payment plans and instalments with due dates	Missing	One <code>dueDate</code> per invoice.
F11	Reissue, credit note, adjustment	Partial	Discard is DRAFT only; a wrong ISSUED invoice needs write-off plus a new invoice.

Collection

#	Workflow	Status	Evidence and gap
F12	Record receipt with allocation (first-fit or explicit), surplus to credit	Implemented	Row locks, split and credit-xor-balance DB checks. Since revised (b21f6f2): a DRAFT invoice is refused by name rather than flipped to PART_PAID, and the receipt is always POSTED.
F13	Post a receipt saved as draft	Missing, by decision	postNow: false creates a DRAFT receipt (receipts/route.ts:94, 499-503); no post route exists; allocate and void both require POSTED. A draft receipt is stuck forever. Closed in b21f6f2 by removal rather than by a post route: postNow is gone and a receipt is always POSTED, because a counter receipt posts when the money is taken and no screen ever asked for anything else. The workflow does not exist to be broken now.
F14	Allocate later, void with reason	Implemented	Void ignores FiscalReceipt state (see §5 B4). Fixed in b21f6f2 : a receipt whose fiscal row is SUCCESS or PENDING is refused, and the refusal names the instrument that would do the job — a fiscal credit note, which the product does not yet issue.
F15	Fiscalise via ZIMRA, replay un-landed receipts	Implemented	Feature-flagged; never blocks payment.
F16	Receipt printed at the counter	Partial	Print is a row verb after the fact; the receipt dialog has no "print on save".
F17	Online payment from the parent portal, reconciled to receipts	Missing	S-7.3, S-7.4.
F18	Cash-up, till close, deposit slip, bank reconciliation of fee receipts	Missing	BankReconciliation exists in accounting but has no link to SchoolFeeReceipt .

#	Workflow	Status	Evidence and gap
F19	Payment method configuration (accounts, merchant codes)	Missing	Four enum values; nothing to tell a parent where to pay. Still missing, and now visible from the other side: the parent portal's "how to pay" sheet was built during the implementation and can source only the amount, the due date and the invoice number to quote, because the tenant record holds no bank or mobile-money detail to put beside them (B16).

Relief and refunds

#	Workflow	Status	Evidence and gap
F20	Waiver create, edit, discard, approve, apply, reject, reverse	Implemented	A waiver can be created directly as APPROVED or APPLIED; apply stamps the caller as approver when none exists; /finance/waivers/[id]/approve is a re-export of apply. No segregation of duties. Fixed in b21f6f2: status is gone from the create schema, so a waiver is born a draft; approval is a PATCH by whoever signs it off; applying an unapproved waiver is refused before the first write; and applying one whose approver is also its author takes a school administrator, because in a small office there may be nobody else and the head answering for it is the school's own decision. The approve re-export is gone. The dialog that raised a waiver was not revisited: it still offers "Approve it now", which the route now strips rather than honours or refuses (B17).
F21	Refund request from credit or receipt surplus, pay, cancel	Implemented	GL posted after commit.
F22	Transport and library charges flowing into invoices	Missing	Transport billing "reported, never posted" (lib/schools/transport.ts:255-262); library fines computed, never billed.

Chasing and reporting

#	Workflow	Status	Evidence and gap
F23	Fees by year group, class finance view	Implemented	
F24	Arrears with ageing, collections, export CSV and PDF	Implemented	Three surfaces compute ageing differently (see UI audit). There were four, and they did not agree; see the correction at B10. Fixed in <code>f6556ee: lib/schools/ageing.ts</code> holds the boundaries, the labels and the reasoning, one strip draws them, the server's report builder asks the same service, and the export's last column is labelled for the ninety days it has always held.
F25	Send reminders to families in arrears	Partial	Finance overview button enabled by <code>fees:create</code> , route needs <code>reports:create (notices/route.ts:121)</code> . Arrears page disables it. Delivery is in-app only; families without a portal account are counted as <code>withoutAccount</code> and not reached. The route is fixed (<code>e5b76a5</code>): it takes <code>notify-families</code> , which the bursar, registrar, HOD and a class teacher hold, and the finance overview button now sends. The arrears page was not repointed and still asks <code>schools.reports.create</code> , so the screen built for this job is the one screen the bursar still cannot send from (B13). Delivery is still in-app only, and a family with no claimed account is still counted and dropped.

#	Workflow	Status	Evidence and gap
F26	Answer a parent's fee question in the office inbox	Implemented	This finding was wrong. It read as a permission check refusing the bursar a reply the office had. There was no reply on <code>POST /messages</code> for anybody: the route accepted assign and close, the 403 fired before the body was read, and the screen — which headed a column "Needs a reply" — carried no composer at all. Replies could be written from the staff portal and nowhere else, so the office's choices were to close a fee thread or hand it to a teacher who cannot see the ledger. The gap was every office role, not the bursar. Fixed in <code>d26c31e</code> : answering is a <code>reply</code> grant held by the head, registrar, bursar and HOD; triage stays on <code>create</code> , because deciding who answers a family is not the same act as answering them; and the composer is hidden for a reader without the grant or on a closed thread rather than shown and then refused.
F27	Dunning schedule, automatic reminders before and after due date	Missing	S-9.3 <code>todo</code> .
F28	Statements per family or per class as a batch	Missing	Statement source exists; only a single-student print.
F29	Results or portal hold for non-payment (fee-threshold gate)	Missing	Promised in <code>schools.md</code> §4.5 and the market gameplan; no story.
F30	Collections target and daily takings	Missing	Reports show collections vs billed; no target, no today figure.

#	Workflow	Status	Evidence and gap
F31	Portal invites to guardians and students	Implemented	Manual link copy; guardians without email are skipped. Both unchanged. The implementation found a third fault the audit had not: because the person issuing the invite chooses the address it is sent to, naming an address that already had a user let the claimer set that user's password, name and role — a teacher's login included. Claiming now opens an account and never adopts one, refused both in the route and in the transaction (1f1501c , B15).
F32	Who changed what a family owed	Implemented	Six fee actions write PlatformAuditEvent in-transaction.
F33	Multi-currency invoicing and receipting	Implemented	Rate per document date; allocation refused across currencies.

Counts as audited: 15 implemented (2 with control gaps), 4 partial, 3 broken, 11 missing. As revised: 16 implemented and none carrying a control gap, 5 partial, none broken, 12 missing — one of which, F13, is missing because the path it described was deleted rather than repaired, not because it was never built.

4. Benchmark gap

Graded against ../reference/k12-benchmark.md §9.

Capability	Benchmark norm	Huchu today	Priority
Fee structures, bulk invoicing, receipts, credits, refunds, write-off	Standard	Present, strong	Done
GL posting and fiscalisation	Regional necessity	Present	Done
Online payment and reconciliation	Standard	Absent	P0
Reminders and dunning by SMS or WhatsApp	Standard	Absent; in-app only. The bursar can now send one, from the finance overview	P0
Payment plans and instalments	Standard	Absent	P1
Sibling, staff and early-payment discounts as rules	Standard	Absent	P1
Scholarship and sponsor (BEAM, church, company) billing	Regional necessity	Absent	P1
Cash-up and bank reconciliation of receipts	Standard	Absent	P1

Capability	Benchmark norm	Huchu today	Priority
Statement runs and family statements	Standard	Single print only	P1
Fee holds on results	Common in the region	Absent	P2 (policy)
Tuck shop and uniform POS with fiscalisation	Sold in the add-on copy	Absent from the pack (retail POS exists separately)	P2
Offline receipting	Sold on the site	Absent	P1
Bursar dashboard with today's takings and collection rate	Standard	Absent (year-group picker)	P1

5. Bugs and risks

#	Finding	Location	Severity
B1	Bursar cannot send reminders or reply in the office inbox: <code>reports:create</code> is SCHOOL_ADMIN only. Half of this was right. The reminder half was: <code>POST /notices</code> did want <code>reports:create</code> , and it now takes <code>notify-families</code> , which the bursar holds (<code>c4689af</code> , <code>e5b76a5</code>). The inbox half was wrong about the mechanism and understated the gap — there was no reply action on <code>POST /messages</code> for anybody, so the 403 fired before the body was read and the refusal reached every office role, not the bursar alone. Fixed in <code>d26c31e</code> ; see F26. The reminder half has a tail of its own at B13.	<code>fees-grade-picker.tsx:184</code> vs <code>notices/route.ts:121</code> ; <code>messages/route.ts:83</code> ; <code>personas.ts:106-124</code>	High
B2	Draft receipt can never be posted, allocated or voided. Fixed in <code>b21f6f2</code> by deleting the state rather than adding the route: <code>postNow</code> is gone and a receipt is always POSTED, because the counter case is the only case any screen ever asked for.	<code>receipts/route.ts:94</code> , <code>499-503</code> ; <code>allocate/route.ts:111</code> ; <code>void/route.ts:43</code>	High

#	Finding	Location	Severity
B3	Waiver approval has no segregation of duties; one bursar grants, approves and applies a scholarship. Fixed in <code>b21f6f2: status</code> is gone from the create schema, so a waiver is born a draft whatever the caller sends; the <code>approvedById ?? session.user.id</code> fallback that invented an approver is gone; applying an unapproved waiver is refused before the first write; and applying one approved by its own author takes a school administrator. The <code>approve</code> route that was a re-export of <code>apply</code> is deleted — approval is a PATCH to the waiver.	<code>waivers/route.ts:35, 197-221; waivers/[id]/apply/route.ts:138</code>	High (control)
B4	Voiding a receipt never checks <code>FiscalReceipt</code>; a receipt already submitted to ZIMRA can be voided locally with no credit note or reversal. Fixed in <code>b21f6f2: SUCCESS</code> and <code>PENDING</code> are both refused — <code>PENDING</code> because the signed row is already holding its counter and its global number while replay resends it — and the refusal names the fiscal credit note as the instrument that would do the job, which the product still does not issue. A <code>FAILED</code> attempt never landed and a receipt with no fiscal row was never sent, so both stay voidable.	<code>receipts/[id]/void/route.ts</code>	High (compliance)

#	Finding	Location	Severity
B5	GL posting after the transaction for receipts, void, write-off and refund pay; a failed posting is <code>console.error</code>ed, no journal id or status stored on the document, no retry. Money can be received with no ledger entry and no list of unposted documents. Half fixed in <code>44c4460</code>: each fee document now carries <code>accountingStatus</code>, <code>journalEntryId</code> and a failure reason, with a <code>companyId</code>-scoped index so "what has not posted" is a cheap question, and the backfilling migration matched historical rows against the journal entry each would have written rather than marking them fine — which is how 120 issued invoices on the seeded tenant turned out to have no ledger entry at all. The emit is still after the commit, deliberately. What is still missing is anyone to ask the question: no screen reads the field, there is no unposted filter and no replay endpoint (B14).	<code>receipts/route.ts:433, 588; void/route.ts:37, 118; write-off/route.ts:52, 92; pay/route.ts:56, 119</code>	High
B6	DRAFT invoices accept allocations, waivers and write-offs, flipping to PART_PAID or PAID without ever being ISSUED, so the receivable is posted without the issue journal. Fixed in <code>b21f6f2</code>, in all three paths: allocation refuses a draft by invoice number, a waiver wants an issued bill and says so, and a draft is discarded rather than written off.	<code>_helpers.ts:630; apply/route.ts:79, 98; write-off/route.ts:40-45</code>	Medium

#	Finding	Location	Severity
B7	PATCH /guardian-links/[id] (consent flags, primary) has no persona check; BURSAR, via <code>canViewAnyPortalSubject</code> , can change which parent receives results. Fixed in <code>e5b76a5</code> : it takes <code>schoools.students.edit</code> , the registrar's grant, which the bursar does not hold. The route-guard test now walks every exported mutating method rather than searching a file for a marker, which is how this one stayed unguarded (<code>4fce301</code>).	<code>guardian-links/[id]/route.ts:46-110</code>	Medium
B8	Opening-balance import requires a role no bursar has. Fixed in <code>50df723</code> : each entity type is checked against the one grant it belongs to. Loading the roll is registrar work; loading fee structures and opening balances is the bursar's, and asking both left only <code>SCHOOL_ADMIN</code> able to do either.	<code>imports/_guard.ts:19-24</code>	Medium
B9	Fee structure creation is not audited while edit, activate and archive are. Fixed in <code>b21f6f2</code> : <code>schoools.fee.structure.created</code> is written on the transaction client alongside the create, so no row describes a structure that rolled back and none is lost to a failed commit.	<code>fees/structures/route.ts</code>	Low

#	Finding	Location	Severity
B10	Three separately coded ageing computations with different bucket sets and labels on the finance overview, the reports page and the dashboard. On the seeded tenant all three agree; the older screenshot set showed them disagreeing. A maintenance risk rather than a live defect. This finding was wrong on both counts, and the runtime pass made it worse by recording the surfaces as agreeing. There were four computations, not three: the server's own report builder in <code>lib/schools/reports.ts</code> was the fourth. And they do not agree in general. The finance overview's SQL called a bill Current only while its due date was still in the future, and a due date is stored at midnight, so a bill due today fell out of Current the moment the clock passed midnight while the arrears report, counting whole days, still called it Current — the same money, two answers, on two screens a bursar reads side by side. The CSV and PDF export labelled its last column "120+ days" for a column that has always held everything past ninety. It was a live defect and should have been recorded as one. Fixed in <code>f6556ee: lib/schools/ageing.ts</code> holds the boundaries, the labels and the reasoning, the screens share one strip, the report builder asks the same service, and the SQL was shifted to match.	<code>fees-grade-picker.tsx:188-192; schools-reports-enhanced-content.tsx:1164-1168; schools-dashboard-content.tsx:96-104, 1212-1216; lib/schools/reports.ts</code>	Medium, and it should have been High

#	Finding	Location	Severity
B11	Parent-facing invoice, receipt and statement downloads always fail for the PARENT role, so the "phone call the bursar does not have to take" still happens. Fixed in 1f1501c : a portal caller is judged by record rather than by role — the route resolves which pupil the document is about and asks whether this guardian is linked to that pupil with the consent flag the document needs, financials for money and academic results for marks. The branch covers only the four documents a portal hands out; class lists and transfer letters stay with the office. Every portal refusal answers alike and with a 404, so a recordId cannot be walked to learn which documents exist, at the deliberate cost of the wording that told a parent their link withholds financials.	app/api/documents/renderer/route.ts:101-108	High (see parent audit)
B12	No browser test records a receipt, issues an invoice, applies a waiver or voids; e2e is a page-render sweep. Still open. The money path did gain unit coverage for everything B2 to B6 closed — waiver segregation, a draft bill taking neither relief nor payment, refusing to void a receipt ZIMRA has seen, and the posting status — and guard coverage is now checked per exported method rather than per file (4fce301). No browser test records a receipt.	e2e/schools-back-office-suite.spec.ts	Medium

#	Finding	Location	Severity
B13	The arrears page still gates "Remind the N" and the row's Remind verb on <code>schools.reports</code> create, which the bursar does not hold, while the route behind them now takes <code>notify-families</code> , which she does. The two screens have swapped places: the finance overview, which the audit found enabled and refused, now works, and the arrears page, which the audit found correctly disabled, is disabled for a permission that is no longer the one being enforced. The tooltip still reads "Writing to families is the head's to do." Recorded during implementation; open.	<code>reports-arrears-content.tsx:409, 426, 445-452</code> vs <code>notices/route.ts:125</code>	High
B14	A fee document's posting outcome is recorded and nothing reads it back. There is no unposted filter on the ledger, no screen that shows a <code>ccountingStatus</code> at all, and no replay endpoint, so the 120 issued invoices the backfill found with no ledger entry exist as a column in the database and nowhere a bursar can see. Recorded during implementation; open.	<code>lib/schools/fee-posting-status.ts</code> ; <code>prisma/schema.prisma:7649-7687</code>	High
B15	The portal claim flow could take over an existing account. The person issuing an invite chooses the address it is sent to, so naming an address that already had a user let the claimer set that user's password, name and role — a teacher's login included. Recorded during implementation. Fixed in <code>1f1501c</code> : claiming opens an account and never adopts one, refused both in the route and in the transaction that did the overwriting, because the service is the part that was actually unsafe and should not depend on its only caller to stay that way.	<code>lib/schools/portal-invites.ts</code> ; <code>app/api/public/schools/claim/[token]/route.ts</code>	High

#	Finding	Location	Severity
B16	The tenant record carries no bank or mobile-money merchant details, so nothing in the product can tell a family where to send money. Recorded during implementation, when the parent portal's "how to pay" sheet was built and could source only the amount, the due date and the invoice number to quote as a reference. This is F19 seen from the family's end, and it is the cheapest half of the payments problem. Open.	parent-how-to-pay-sheet.tsx:13-26 ; F19	Medium
B17	The waiver dialog still offers "Approve it now, rather than leaving it a draft". createSchema no longer has a status field, so Zod strips it, the route answers 201 and the waiver is a draft: the box is neither honoured nor refused, and nothing tells the bursar it did nothing. The control the checkbox used to defeat is correctly closed (B3); the checkbox that defeated it was left on the screen. Recorded during implementation; open.	fee-dialogs.tsx:377, 406, 496-499 vs waivers/route.ts:39-48	Medium

6. Proposed edits

Numbers 2 to 9 are done, 1 in part and 11 in part, in the commits named against the findings in §5; 10 stands untouched. What was built differed from what was proposed in three places, and each difference is recorded under its item.

- 1. Give the bursar the verbs the job needs (B1).** Add a `notify-families` action on `schools.fees` (or grant `schools.reports create` scoped to fee notices) and guard the arrears and finance reminder path on it. Grant BURSAR `schools.messages reply` (new resource) so fee threads can be answered. Gate the finance overview button on the same check the arrears page uses. Done for the verbs, on `schools.reports` rather than `schools.fees` or a new resource: the actions are `notify-families` and `reply`, and triage — assigning a thread and ending it — deliberately stays on `create`. Not done for the last sentence, and it now matters in the opposite direction: the arrears page is the screen still asking the wrong question (B13).
- 2. Add `POST /fees/receipts/[id]/post` (B2)** or drop `postNow` entirely; the counter case is always post-now. Done by the second route.
- 3. Segregation of duties on waivers (B3).** Require `approve` by `SCHOOL_ADMIN` (or a second bursar) before `apply`; refuse `approvedById === createdById`; remove the create-as-APPLIED path. Keep the audit rows. Done, with one softening: `approvedById === createdById` is refused for everyone except a

school administrator, because a small office may have nobody else and the head answering for it is the school's own decision.

4. **Fiscal-aware void (B4).** Refuse void when a linked `FiscalReceipt` is submitted, or route it through a fiscal credit note. Done by the first route; the credit note is still not an instrument the product issues.
5. **Persist posting outcome (B5).** Store `journalEntryId` and `accountingStatus` on `SchoolFeeReceipt`, `SchoolFeeInvoice`, `SchoolFeeRefund` (as `FiscalReceipt` does), add a "not posted" filter and a replay endpoint, or emit inside the transaction. Done for the storage, with a backfill; the filter and the replay endpoint are not built (B14).
6. **Restrict allocation, waiver and write-off to ISSUED and PART_PAID (B6).** Done, PAID included for allocation, because a fully settled bill is still what a parent pays twice against and the second payment is carried as credit rather than refused.
7. **Persona check on guardian-link PATCH (B7)** and split the import guard per entity (B8). Both done.
8. **One ageing service (B10)** with one bucket definition and one component, consumed by all three surfaces. Done, by four surfaces: the server's report builder was the one the finding missed.
9. **Fix the parent downloads (B11)** with a guardian-link check in the render path. Done, with every portal refusal answering alike and with a 404 so a `recordId` cannot be walked.
10. **Browser tests for the money path (B12):** record receipt, issue, bulk generate, waiver apply, refund pay, void, in `e2e/`. Not done. The same paths gained unit tests instead, which is a different thing.
11. **Audit structure creation (B9)** and update the roadmap rows for S-0.6 (posting is after commit) and S-5.1 (receipt and statement have no page). The audit row is written; the roadmap rows are not.

7. Proposed restructuring

- **Make `/schools/finance` the bursar's home and a dashboard**, not a year-group picker: collected vs billed hero, today's takings, drafts not issued, receipts not fiscalised, refunds awaiting payment, ageing strip, longest overdue with verbs, recent receipts. "Record receipt" as the one primary in the bar. The year-group table stays below. Not done; the screen is the year-group picker it was. What did change under it is that the rail now lands the bursar here, and the Send reminders button on it sends.
- **One ledger, one status vocabulary.** The ledger's Draft, Issued, Part paid, Paid, Written off, Voided everywhere; "Overdue" becomes a filter, not a status. Half done (`c65e58a`): "Overdue" is a filter on the ledger — due date past, balance above zero — and no longer sits among the states. The students list still badges a pupil's fee standing Paid, Partial, Overdue.
- **Family as a unit.** Receipts, statements and reminders are per family in practice. Introduce a household view (guardian with billing responsibility, all children) so one receipt can settle several children and one statement covers the family. Not done.
- **Retire the fees phase 3 spec** and fold its surviving intent into the roadmap. Not done.
- **Separate relief from billing.** Discounts and scholarships become schemes applied at invoice generation; waivers remain the exception path for one-off relief with approval. Not done, though the second half of the sentence is now true of the waiver: it is an exception path with a real approval, taken by somebody other than the person who raised it.

8. Proposed new workflows and features

Each entry names the model and route it needs. None of these has been built, and the implementation work did not attempt them. Still absent as this document is revised: any channel out of the building — no email, no SMS, no WhatsApp — so a reminder reaches only a family with a claimed portal account, and no password reset or self-service for the accounts that are claimed; payment inside the portal, and no bank or mobile-money merchant details on the tenant record, so the family's "how to pay" sheet can show only the amount, the due date and the invoice number to quote (B16); offline receipting; payment plans; schemes; a cash-up or bank reconciliation; a statement run; and a dunning schedule. Bulk issue, print, activate and approve have no endpoint either, and the ledger ships only the one bulk verb the API can perform in a single call rather than fanning out browser requests to fake the rest.

1. **Online payments (S-7.3, S-7.4).** `SchoolFeePayment` intent (invoice ids, amount, method, provider ref, status INITIATED / PENDING / PAID / FAILED) on the `lib/payments/` seam; on PAID create the receipt through the existing allocation and posting; bursar screen "Portal payments" with method filter and unreconciled list. Providers: Paynow (EcoCash, OneMoney), card, bank transfer with proof upload and bursar confirmation.
2. **Reminders with a channel (S-9.3).** Invoice issued, due in 7 days, overdue at 7, 30 and 60 days; per family; in-app plus SMS or WhatsApp; bursar sees delivery status and cost. Templates per school.
3. **Payment plans.** `SchoolFeePaymentPlan` with instalments (amount, due date) against an invoice; reminders per instalment; arrears computed per instalment.
4. **Discount and scholarship schemes.** `SchoolFeeScheme` (type SIBLING, STAFF, EARLY_PAYMENT, SCHOLARSHIP, SPONSOR; rule; sponsor party) applied at bulk generation as invoice lines or credit notes; sponsor billing produces an invoice to the sponsor.
5. **Cash-up and banking.** `SchoolCashSession` per cashier per day (opening float, receipts by method, count, variance, deposit slip); link receipts to bank statement lines through the existing `BankReconciliation`.
6. **Statement run.** Batch statements per class or whole school at term start and mid-term, delivered to the portal and by WhatsApp or email, using `schools.fee.statement`.
7. **Transport and library billing into invoices.** Rider fees as structure lines; fines as a "Library" line on the next invoice or a separate small invoice.
8. **Fee hold policy** (optional per school): results or report card withheld below a payment threshold, with the head's override and an audit row.
9. **Offline receipting (S-8.2)** through the outbox with an idempotency key on receipt number.
10. **Tuck shop and uniform sales:** either wire the retail POS to school fee customers with fiscalisation, or remove the add-on copy.

9. Open decisions

- Who approves a waiver: the head, a second bursar, or a configurable threshold? Partly settled: a second person, whoever they are, and a school administrator may be both approver and applier because a small office may have nobody else. A configurable threshold is still open.

- Which payment provider for fees, and who bears the transaction fee? Open, and now blocking a smaller thing as well: until the tenant record carries bank and mobile-money details, the portal cannot tell a family where to send money by any means at all (B16).
- Is a receipt ever saved as draft? If not, delete the path. Settled: no, and the path is deleted.
- Should the school see accounting screens, or stay headless? The Premier band says one thing and the scope trim another.
- Family or student as the billing unit going forward.